

GraphTheorySymbol: L^AT_EX Commands with Symbols for Graph Theory

Laurent Frédéric Bernard François Lyaudet
Website:<https://lyaudet.eu/laurent/>
Email:Laurent.Lyaudet@gmail.com

v1.5.1 2026/09/13

Abstract

This documentation is for L^AT_EX package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

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1 Introduction

This documentation is for L^AT_EX package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

This package is licenced under Lesser GNU General Public License version 3 or later (LGPLv3+). It means you are free do anything with it, but if you do improve the code of the symbols that we provide, please do share these improvements in a similar way. There is no restriction on other graph theory symbols/code, even if it would be a good fit for inclusion in this package. There is also no need to share in a similar way what you build with this package. The text of the licenses is available at:

<https://www.gnu.org/licenses/>.

The git repository of this source code is also available at:

<https://github.com/LLyaudet/GraphTheorySymbol/>.

Clone it with this URL:

[git@github.com:LLyaudet/GraphTheorySymbol.git](https://github.com/LLyaudet/GraphTheorySymbol.git).

This package is available on CTAN:

<https://ctan.org/pkg/graph-theory-symbol>.

This is our first $\text{\LaTeX}$ package. It may be that our macros are quite slow and that a document making extensive use of these macros will be slow to compile. If you know how to produce better $\text{\LaTeX}$ code, please submit a pull request on GitHub.

2 Usage

2.1 Memo or for the impatient

Add in your preamble: `\usepackage{graph-theory-symbol}`.

For clarity, below we only show long macro names and short macro names; see Detailed usage section for intermediate length macro names.

Long name	Short name	Result	In context
<code>\GTSadjacency/</code>	<code>\GTSa/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSadjacencyRelation/</code>	<code>\GTSaR/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSunadjacency/</code>	<code>\GTSu/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSunadjacencyRelation/</code>	<code>\GTSuR/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSdirectedAdjacencyUp/</code>	<code>\GTSdAU/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSdirectedAdjacencyUpRelation/</code>	<code>\GTSdAUR/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSnegatedDirectedAdjacencyUp/</code>	<code>\GTSnDAU/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSnegatedDirectedAdjacencyUpRelation/</code>	<code>\GTSnDAUR/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSdirectedAdjacencyDown/</code>	<code>\GTSdAD/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSdirectedAdjacencyDownRelation/</code>	<code>\GTSdADR/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSnegatedDirectedAdjacencyDown/</code>	<code>\GTSnDAD/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSnegatedDirectedAdjacencyDownRelation/</code>	<code>\GTSnDADR/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSincidence/</code>	<code>\GTSi/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSincidenceRelation/</code>	<code>\GTSiR/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSnegatedIncidence/</code>	<code>\GTSnI/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSnegatedIncidenceRelation/</code>	<code>\GTSnIR/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSoutIncidence/</code>	<code>\GTSOI/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSoutIncidenceRelation/</code>	<code>\GTSOIR/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSnegatedOutIncidence/</code>	<code>\GTSnOI/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSnegatedOutIncidenceRelation/</code>	<code>\GTSnOIR/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSinIncidence/</code>	<code>\GTSiI/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSinIncidenceRelation/</code>	<code>\GTSiIR/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSnegatedInIncidence/</code>	<code>\GTSnII/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSnegatedInIncidenceRelation/</code>	<code>\GTSnIIR/</code>	$\mathbb{G}$	$v \mathbb{G} e$

Long name	Short name	Result	In context
<code>\GTSadjacencyMonochrome/</code>	<code>\GTSaM/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSadjacencyMonochromeRelation/</code>	<code>\GTSaMR/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSunadjacencyMonochrome/</code>	<code>\GTSuM/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSunadjacencyMonochromeRelation/</code>	<code>\GTSuMR/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSdirectedAdjacencyUpMonochrome/</code>	<code>\GTSdAUM/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSdirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSdAUMR/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSnegatedDirectedAdjacencyUpMonochrome/</code>	<code>\GTSnDAUM/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSnegatedDirectedAdjacencyUpMonochromeRelation/</code>	<code>\GTSnDAUMR/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSdirectedAdjacencyDownMonochrome/</code>	<code>\GTSdADM/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSdirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSdADMR/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSnegatedDirectedAdjacencyDownMonochrome/</code>	<code>\GTSnDADM/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSnegatedDirectedAdjacencyDownMonochromeRelation/</code>	<code>\GTSnDADMR/</code>	$\mathbb{G}$	$u \mathbb{G} v$
<code>\GTSincidenceMonochrome/</code>	<code>\GTSiM/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSincidenceMonochromeRelation/</code>	<code>\GTSiMR/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSnegatedIncidenceMonochrome/</code>	<code>\GTSnIM/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSnegatedIncidenceMonochromeRelation/</code>	<code>\GTSnIMR/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSoutIncidenceMonochrome/</code>	<code>\GTSOIM/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSoutIncidenceMonochromeRelation/</code>	<code>\GTSOIMR/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSnegatedOutIncidenceMonochrome/</code>	<code>\GTSnOIM/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSnegatedOutIncidenceMonochromeRelation/</code>	<code>\GTSnOIMR/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSinIncidenceMonochrome/</code>	<code>\GTSiIM/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSinIncidenceMonochromeRelation/</code>	<code>\GTSiIMR/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSnegatedInIncidenceMonochrome/</code>	<code>\GTSnIIM/</code>	$\mathbb{G}$	$v \mathbb{G} e$
<code>\GTSnegatedInIncidenceMonochromeRelation/</code>	<code>\GTSnIIMR/</code>	$\mathbb{G}$	$v \mathbb{G} e$

<code>\GTSsetMainColor{blue}</code>			
Long name	Short name	Result	In context
<code>\GTSadjacency/</code>	<code>\GTSa/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSadjacencyRelation/</code>	<code>\GTSaR/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSunadjacency/</code>	<code>\GTSu/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSunadjacencyRelation/</code>	<code>\GTSuR/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSdirectedAdjacencyUp/</code>	<code>\GTSdAU/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSdirectedAdjacencyUpRelation/</code>	<code>\GTSdAUR/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSnegatedDirectedAdjacencyUp/</code>	<code>\GTSnDAU/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSnegatedDirectedAdjacencyUpRelation/</code>	<code>\GTSnDAUR/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSdirectedAdjacencyDown/</code>	<code>\GTSdAD/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSdirectedAdjacencyDownRelation/</code>	<code>\GTSdADR/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSnegatedDirectedAdjacencyDown/</code>	<code>\GTSnDAD/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSnegatedDirectedAdjacencyDownRelation/</code>	<code>\GTSnDADR/</code>		$u \overset{\circ}{\sim} v$
<code>\GTSincidence/</code>	<code>\GTSi/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSincidenceRelation/</code>	<code>\GTSiR/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSnegatedIncidence/</code>	<code>\GTSnI/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSnegatedIncidenceRelation/</code>	<code>\GTSnIR/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSoutIncidence/</code>	<code>\GTSOI/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSoutIncidenceRelation/</code>	<code>\GTSOIR/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSnegatedOutIncidence/</code>	<code>\GTSnOI/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSnegatedOutIncidenceRelation/</code>	<code>\GTSnOIR/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSinIncidence/</code>	<code>\GTSiI/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSinIncidenceRelation/</code>	<code>\GTSiIR/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSnegatedInIncidence/</code>	<code>\GTSnII/</code>		$v \overset{\circ}{\sim} e$
<code>\GTSnegatedInIncidenceRelation/</code>	<code>\GTSnIIR/</code>		$v \overset{\circ}{\sim} e$

Long name	Short name	Result	In context
<code>\GTSmetaNot/</code>	<code>\GTSmN/</code>	$\neg$	$\neg \overset{\circ}{\sim}$
<code>\GTSmetaAnd/</code>	<code>\GTSmA/</code>	$\wedge$	$\overset{\circ}{\sim} \wedge \overset{\circ}{\sim}$
<code>\GTSmetaOr/</code>	<code>\GTSmO/</code>	$\vee$	$\overset{\circ}{\sim} \vee \overset{\circ}{\sim}$

Implicit meta-and via concatenation, no space except around whole relations compositions, no parentheses via higher priority to meta-and, brackets when all possible adjacencies are specified between two vertices:

Tournament adjacency:

$\forall u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

Tournament adjacency with maybe some colored edge also, who knows?:

$\forall u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

Directed graph:

$\forall u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

Oriented graph:

$\forall u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

Prefix notation:

$\exists u, v, u \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} v$.

$\exists v, e, v \overset{\circ}{\sim} \overset{\circ}{\sim} \overset{\circ}{\sim} e$.

Apart from these symbol macros, versatile macros to call them all are available with options:

`\GTSxa[]`, `\GTSxadj[]`, or `\GTSxadjacency[]` for adjacency symbols,

`\GTSxi[]`, `\GTSxinc[]`, or `\GTSxincidence[]` for incidence symbols,

`\GTSxml[]`, or `\GTSxmetallogic[]` for meta-logic symbols,

`\GTSx[]{}` for all symbols

$(\backslash\text{GTSx}[\{a\}] = \backslash\text{GTSxa}[], \backslash\text{GTSx}[\{i\}] = \backslash\text{GTSxi}[], \backslash\text{GTSx}[\{m\}] = \backslash\text{GTSxml}[])$.

See Detailed usage section for full syntax of these macros.

2.2 Detailed usage

First you need to add in your preamble: `\usepackage{graph-theory-symbol}`.

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

2.2.1 General macros

\GTS@newcommand This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{\commandname}` The name of the command to create.

`{\commandcode}` The code of the command to create.

\GTS@newprotectedcommand This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

`{\commandname}` The name of the command to create.

`{\commandcode}` The code of the command to create.

\GTSsetMainColor This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

`{\color}` The color that must be used by GraphTheorySymbol afterward.

\GTSsetSecondColor This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

`{\color}` The color that must be used by GraphTheorySymbol afterward.

2.2.2 Standard adjacency macros

\GTS@a This macro will draw any of the adjacency symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and **\GTS@b**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

`{\mainColor}` If non-empty, this color will be used as main color.

`{\secondColor}` If non-empty, this color will be used as second color.

`{\drawSlash}` Non-empty implies that a negation will be drawn.

`{\drawUpArrow}` Non-empty implies that an up-arrow will be drawn.

`{\drawDownArrow}` Non-empty implies that a down-arrow will be drawn.

\GTS@b This macro builds on top of **\GTS@a** to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and **\GTS@a**: **\GTSxadjacency**, **\GTSxadj**, and **\GTSxa**.

Arguments:

`{\mainColor}` If non-empty, this color will be used as main color.

`{\secondColor}` If non-empty, this color will be used as second color.

`{\drawSlash}` Non-empty implies that a negation will be drawn.

`{\drawUpArrow}` Non-empty implies that an up-arrow will be drawn.

`{\drawDownArrow}` Non-empty implies that a down-arrow will be drawn.

`{\relation}` Non-empty implies that relation spacing will be added.

\GTSxadjacency, **\GTSxadj**, **\GTSxa** Versatile macros for adjacency that can be given options.

Argument:

`[{\optionsList}]` Sets the following options if not empty:

`mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,

`secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,

`monochrome` or `mono` or `mon` or `m` (equivalent to `mC=black,sC=black`),

`negated` or `not` or `n`,

`up` or `u`,

down or d,
relation or rel or r.

\GTSadjacency, \GTSadj, \GTSa This macro will draw an adjacency symbol. A first drawing didn't include the small dots in the middle of the two circles corresponding to the graph vertices. But it was too similar to “spoon” and multimap symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property to remind of the classical textual notation of adjacency relation as $\text{Adj}(.,.)$.

A.\GTSadjacency/A.
or A.\GTSadj/A.
or A.\GTSa/A.
will yield $A \circ A$.
u \GTSadjacency/ v
or u \GTSadj/ v
or u \GTSa/ v
will yield $u \circ v$.
\$u \GTSadjacency/ v\$
or \$u \GTSadj/ v\$
or \$u \GTSa/ v\$
will yield $u \circ v$.

\GTSadjacencyRelation, \GTSadjRel, \GTSaR This macro will draw an adjacency symbol.

\$u\GTSadjacencyRelation/v\$
or \$u\GTSadjRel/v\$
or \$u\GTSaR/v\$
will yield $u \circ v$.

\GTSunadjacency, \GTSunadj, \GTSu This macro will draw an unadjacency symbol.

A.\GTSunadjacency/A.
or A.\GTSunadj/A.
or A.\GTSu/A.
will yield $A \circ A$.
u \GTSunadjacency/ v
or u \GTSunadj/ v
or u \GTSu/ v
will yield $u \circ v$.
\$u \GTSunadjacency/ v\$
or \$u \GTSunadj/ v\$
or \$u \GTSu/ v\$
will yield $u \circ v$.

\GTSunadjacencyRelation, \GTSunadjRel, \GTSuR This macro will draw an unadjacency symbol.

\$u\GTSunadjacencyRelation/v\$
or \$u\GTSunadjRel/v\$
or \$u\GTSuR/v\$
will yield $u \circ v$.

\GTSdirectedAdjacencyUp, \GTSdirAdjUp, \GTSdAU This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

A.\GTSdirectedAdjacencyUp/A.
or A.\GTSdirAdjUp/A.
or A.\GTSdAU/A.
will yield $A \circ A$.
u \GTSdirectedAdjacencyUp/ v
or u \GTSdirAdjUp/ v
or u \GTSdAU/ v
will yield $u \circ v$.
\$u \GTSdirectedAdjacencyUp/ v\$
or \$u \GTSdirAdjUp/ v\$

or $\$u \backslash \text{GTSdAU} / v\$$
will yield $u \overset{\curvearrowright}{\sim} v$.

$\backslash \text{GTSdirectedAdjacencyUpRelation}$, $\backslash \text{GTSdirAdjUpRel}$, $\backslash \text{GTSdAUR}$ This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

$\$u \backslash \text{GTSdirectedAdjacencyUpRelation} / v\$$
or $\$u \backslash \text{GTSdirAdjUpRel} / v\$$
or $\$u \backslash \text{GTSdAUR} / v\$$
will yield $u \overset{\curvearrowright}{\sim} v$.

$\backslash \text{GTSnegatedDirectedAdjacencyUp}$, $\backslash \text{GTSnegDirAdjUp}$, $\backslash \text{GTSnDAU}$ This macro will draw a negated directed adjacency upward symbol.

$A. \backslash \text{GTSnegatedDirectedAdjacencyUp} / A.$
or $A. \backslash \text{GTSnegDirAdjUp} / A.$
or $A. \backslash \text{GTSnDAU} / A.$
will yield $A. \overset{\curvearrowright}{\sim} A.$

$u \backslash \text{GTSnegatedDirectedAdjacencyUp} / v$
or $u \backslash \text{GTSnegDirAdjUp} / v$
or $u \backslash \text{GTSnDAU} / v$
will yield $u \overset{\curvearrowright}{\sim} v$.
 $\$u \backslash \text{GTSnegatedDirectedAdjacencyUp} / v\$$
or $\$u \backslash \text{GTSnegDirAdjUp} / v\$$
or $\$u \backslash \text{GTSnDAU} / v\$$
will yield $u \overset{\curvearrowright}{\sim} v$.

$\backslash \text{GTSnegatedDirectedAdjacencyUpRelation}$, $\backslash \text{GTSnegDirAdjUpRel}$, $\backslash \text{GTSnDAUR}$ This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

$\$u \backslash \text{GTSnegatedDirectedAdjacencyUpRelation} / v\$$
or $\$u \backslash \text{GTSnegDirAdjUpRel} / v\$$
or $\$u \backslash \text{GTSnDAUR} / v\$$
will yield $u \overset{\curvearrowright}{\sim} v$.

$\backslash \text{GTSdirectedAdjacencyDown}$, $\backslash \text{GTSdirAdjDown}$, $\backslash \text{GTSdAD}$ This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

$A. \backslash \text{GTSdirectedAdjacencyDown} / A.$
or $A. \backslash \text{GTSdirAdjDown} / A.$
or $A. \backslash \text{GTSnDAD} / A.$
will yield $A. \overset{\curvearrowleft}{\sim} A.$
 $u \backslash \text{GTSdirectedAdjacencyDown} / v$
or $u \backslash \text{GTSdirAdjDown} / v$
or $u \backslash \text{GTSnDAD} / v$
will yield $u \overset{\curvearrowleft}{\sim} v$.
 $\$u \backslash \text{GTSdirectedAdjacencyDown} / v\$$
or $\$u \backslash \text{GTSdirAdjDown} / v\$$
or $\$u \backslash \text{GTSnDAD} / v\$$
will yield $u \overset{\curvearrowleft}{\sim} v$.

$\backslash \text{GTSdirectedAdjacencyDownRelation}$, $\backslash \text{GTSdirAdjDownRel}$, $\backslash \text{GTSdADR}$ This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

$\$u \backslash \text{GTSdirectedAdjacencyDownRelation} / v\$$
or $\$u \backslash \text{GTSdirAdjDownRel} / v\$$
or $\$u \backslash \text{GTSdADR} / v\$$
will yield $u \overset{\curvearrowleft}{\sim} v$.

$\backslash \text{GTSnegatedDirectedAdjacencyDown}$, $\backslash \text{GTSnegDirAdjDown}$, $\backslash \text{GTSnDAD}$ This macro will draw a negated directed adjacency downward symbol.

$A. \backslash \text{GTSnegatedDirectedAdjacencyDown} / A.$
or $A. \backslash \text{GTSnegDirAdjDown} / A.$
or $A. \backslash \text{GTSnDAD} / A.$

will yield $A \nrightarrow A$.
`u \GTSnegatedDirectedAdjacencyDown/ v`
or `u \GTSnegDirAdjDown/ v`
or `u \GTSnDAD/ v`
will yield $u \nrightarrow v$.
`\$u \GTSnegatedDirectedAdjacencyDown/ v\$`
or `\$u \GTSnegDirAdjDown/ v\$`
or `\$u \GTSnDAD/ v\$`
will yield $u \nrightarrow v$.

`\GTSnegatedDirectedAdjacencyDownRelation`, `\GTSnegDirAdjDownRel`, `\GTSnDADR` This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.
`\$u \GTSnegatedDirectedAdjacencyDownRelation/ v\$`
or `\$u \GTSnegDirAdjDownRel/ v\$`
or `\$u \GTSnDADR/ v\$`
will yield $u \nrightarrow v$.

2.2.3 Standard incidence macros

`\GTS@i` This macro will draw any of the incidence symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and `\GTS@j`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

`{\mainColor}` If non-empty, this color will be used as main color.
`{\secondColor}` If non-empty, this color will be used as second color.
`{\drawSlash}` Non-empty implies that a negation will be drawn.
`{\drawUpArrow}` Non-empty implies that an up-arrow will be drawn.
`{\drawDownArrow}` Non-empty implies that a down-arrow will be drawn.

`\GTS@j` This macro builds on top of `\GTS@i` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@i`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`.

Arguments:

`{\mainColor}` If non-empty, this color will be used as main color.
`{\secondColor}` If non-empty, this color will be used as second color.
`{\drawSlash}` Non-empty implies that a negation will be drawn.
`{\drawUpArrow}` Non-empty implies that an up-arrow will be drawn.
`{\drawDownArrow}` Non-empty implies that a down-arrow will be drawn.
`{\relation}` Non-empty implies that relation spacing will be added.

`\GTSxincidence`, `\GTSxinc`, `\GTSxi` Versatile macros for incidence that can be given options.

Argument:

`[\optionsList]` Sets the following options if not empty:
`mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,
`secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,
`monochrome` or `mono` or `mon` or `m` (equivalent to `mC=black,sC=black`),
`negated` or `not` or `n`,
`in` or `i`,
`out` or `o`,
`relation` or `rel` or `r`.

`\GTSincidence`, `\GTSinc`, `\GTSi` This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

`A.\GTSincidence/A`.
or `A.\GTSinc/A`.
or `A.\GTSi/A`.
will yield $A \downarrow A$.
`v \GTSincidence/ e`

`or v \GTSinc/ e`
`or v \GTSi/ e`
 will yield $v \downarrow e$.
`$v \GTSincidence/ e$`
`or $v \GTSinc/ e$`
`or $v \GTSi/ e$`
 will yield $v \downarrow e$.

`\GTSincidenceRelation, \GTSincRel, \GTSiR` This macro will draw an incidence symbol with relation spacing in math mode.
`$v\GTSincidenceRelation/e$`
`or $v\GTSincRel/e$`
`or $v\GTSiR/e$`
 will yield $v \downarrow e$.

`\GTSnegatedIncidence, \GTSnegInc, \GTSnI` This macro will draw a negated incidence symbol.
`A.\GTSnegatedIncidence/A.`
`or A.\GTSnegInc/A.`
`or A.\GTSnI/A.`
 will yield $A \not\downarrow A$.
`v \GTSnegatedIncidence/ e`
`or v \GTSnegInc/ e`
`or v \GTSnI/ e`
 will yield $v \not\downarrow e$.
`$v \GTSnegatedIncidence/ e$`
`or $v \GTSnegInc/ e$`
`or $v \GTSnI/ e$`
 will yield $v \not\downarrow e$.

`\GTSnegatedIncidenceRelation, \GTSnegIncRel, \GTSnIR` This macro will draw a negated incidence symbol with relation spacing in math mode.
`$v\GTSnegatedIncidenceRelation/e$`
`or $v\GTSnegIncRel/e$`
`or $v\GTSnIR/e$`
 will yield $v \not\downarrow e$.

`\GTSoutIncidence, \GTSoutInc, \GTSoI` This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.
`A.\GTSoutIncidence/A.`
`or A.\GTSoutInc/A.`
`or A.\GTSoI/A.`
 will yield $A \uparrow A$.
`v \GTSoutIncidence/ e`
`or v \GTSoutInc/ e`
`or v \GTSoI/ e`
 will yield $v \uparrow e$.
`$v \GTSoutIncidence/ e$`
`or $v \GTSoutInc/ e$`
`or $v \GTSoI/ e$`
 will yield $v \uparrow e$.

`\GTSoutIncidenceRelation, \GTSoutIncRel, \GTSoIR` This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.
`$v\GTSoutIncidenceRelation/e$`
`or $v\GTSoutIncRel/e$`
`or $v\GTSoIR/e$`
 will yield $v \uparrow e$.

\GTSnegatedOutIncidence, \GTSnegOutInc, \GTSnOI This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

A.\GTSnegatedOutIncidence/A.

or A.\GTSnegOutInc/A.

or A.\GTSnOI/A.

will yield $A \overset{\circ}{\leftarrow} A$.

v \GTSnegatedOutIncidence/ e

or v \GTSnegOutInc/ e

or v \GTSnOI/ e

will yield $v \overset{\circ}{\leftarrow} e$.

\$v \GTSnegatedOutIncidence/ e\$

or \$v \GTSnegOutInc/ e\$

or \$v \GTSnOI/ e\$

will yield $v \overset{\circ}{\leftarrow} e$.

\GTSnegatedOutIncidenceRelation, \GTSnegOutIncRel, \GTSnOIR This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

\$v\GTSnegatedOutIncidenceRelation/e\$

or \$v\GTSnegOutIncRel/e\$

or \$v\GTSnOIR/e\$

will yield $v \overset{\circ}{\leftarrow} e$.

\GTSinIncidence, \GTSinInc, \GTSiI This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

A.\GTSinIncidence/A.

or A.\GTSinInc/A.

or A.\GTSiI/A.

will yield $A \overset{\circ}{\rightarrow} A$.

v \GTSinIncidence/ e

or v \GTSinInc/ e

or v \GTSiI/ e

will yield $v \overset{\circ}{\rightarrow} e$.

\$v \GTSinIncidence/ e\$

or \$v \GTSinInc/ e\$

or \$v \GTSiI/ e\$

will yield $v \overset{\circ}{\rightarrow} e$.

\GTSinIncidenceRelation, \GTSinIncRel, \GTSiIR This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

\$v\GTSinIncidenceRelation/e\$

or \$v\GTSinIncRel/e\$

or \$v\GTSiIR/e\$

will yield $v \overset{\circ}{\rightarrow} e$.

\GTSnegatedInIncidence, \GTSnegInInc, \GTSnII This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

A.\GTSnegatedInIncidence/A.

or A.\GTSnegInInc/A.

or A.\GTSnII/A.

will yield $A \overset{\circ}{\leftarrow} A$.

v \GTSnegatedInIncidence/ e

or v \GTSnegInInc/ e

or v \GTSnII/ e

will yield $v \overset{\circ}{\leftarrow} e$.

\$v \GTSnegatedInIncidence/ e\$

or \$v \GTSnegInInc/ e\$

or \$v \GTSnII/ e\$

will yield $v \overset{\circ}{\leftarrow} e$.

\GTSnegatedInIncidenceRelation, \GTSnegInIncRel, \GTSnIIR This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.
 $\$v\backslash\text{GTSnegatedInIncidenceRelation}/e\$$
or $\$v\backslash\text{GTSnegInIncRel}/e\$$
or $\$v\backslash\text{GTSnIIR}/e\$$
will yield $v \nrightarrow e$.

2.2.4 Monochrome adjacency macros

\GTSadjacencyMonochrome, \GTSadjMono, \GTSadjMon, \GTSaM This macro will draw a monochrome adjacency symbol.
 $A.\backslash\text{GTSadjacencyMonochrome}/A.$
or $A.\backslash\text{GTSadjMono}/A.$
or $A.\backslash\text{GTSadjMon}/A.$
or $A.\backslash\text{GTSaM}/A.$
will yield $A \circ A$.
 $u \backslash\text{GTSadjacencyMonochrome}/ v$
or $u \backslash\text{GTSadjMono}/ v$
or $u \backslash\text{GTSadjMon}/ v$
or $u \backslash\text{GTSaM}/ v$
will yield $u \circ v$.
 $\$u \backslash\text{GTSadjacencyMonochrome}/ v\$$
or $\$u \backslash\text{GTSadjMono}/ v\$$
or $\$u \backslash\text{GTSadjMon}/ v\$$
or $\$u \backslash\text{GTSaM}/ v\$$
will yield $u \circ v$.

\GTSadjacencyMonochromeRelation, \GTSadjMonoRel, \GTSadjMonRel, \GTSaMR This macro will draw a monochrome adjacency symbol with relation spacing in math mode.
 $\$u\backslash\text{GTSadjacencyMonochromeRelation}/v\$$
or $\$u\backslash\text{GTSadjMonoRel}/v\$$
or $\$u\backslash\text{GTSadjMonRel}/v\$$
or $\$u\backslash\text{GTSaMR}/v\$$
will yield $u \circ v$.

\GTSunadjacencyMonochrome, \GTSunadjMono, \GTSunadjMon, \GTSuM This macro will draw a monochrome unadjacency symbol.
 $A.\backslash\text{GTSunadjacencyMonochrome}/A.$
or $A.\backslash\text{GTSunadjMono}/A.$
or $A.\backslash\text{GTSunadjMon}/A.$
or $A.\backslash\text{GTSuM}/A.$
will yield $A \circ A$.
 $u \backslash\text{GTSunadjacencyMonochrome}/ v$
or $u \backslash\text{GTSunadjMono}/ v$
or $u \backslash\text{GTSunadjMon}/ v$
or $u \backslash\text{GTSuM}/ v$
will yield $u \circ v$.
 $\$u \backslash\text{GTSunadjacencyMonochrome}/ v\$$
or $\$u \backslash\text{GTSunadjMono}/ v\$$
or $\$u \backslash\text{GTSunadjMon}/ v\$$
or $\$u \backslash\text{GTSuM}/ v\$$
will yield $u \circ v$.

\GTSunadjacencyMonochromeRelation, \GTSunadjMonoRel, \GTSunadjMonRel, \GTSuMR
This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.
 $\$u\backslash\text{GTSunadjacencyMonochromeRelation}/v\$$
or $\$u\backslash\text{GTSunadjMonoRel}/v\$$
or $\$u\backslash\text{GTSunadjMonRel}/v\$$
or $\$u\backslash\text{GTSuMR}/v\$$
will yield $u \circ v$.

\GTSdirectedAdjacencyUpMonochrome, \GTSdirAdjUpMono, \GTSdirAdjUpMon, \GTSdAUM This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

A.\GTSdirectedAdjacencyUpMonochrome/A.

or A.\GTSdirAdjUpMono/A.

or A.\GTSdirAdjUpMon/A.

or A.\GTSdAUM/A.

will yield $A \overset{\circ}{\curvearrowright} A$.

u \GTSdirectedAdjacencyUpMonochrome/ v

or u \GTSdirAdjUpMono/ v

or u \GTSdirAdjUpMon/ v

or u \GTSdAUM/ v

will yield $u \overset{\circ}{\curvearrowright} v$.

\$u \GTSdirectedAdjacencyUpMonochrome/ v\$

or \$u \GTSdirAdjUpMono/ v\$

or \$u \GTSdirAdjUpMon/ v\$

or \$u \GTSdAUM/ v\$

will yield $u \overset{\circ}{\curvearrowright} v$.

\GTSdirectedAdjacencyUpMonochromeRelation, \GTSdirAdjUpMonoRel, \GTSdirAdjUpMonRel, \GTSdAUMR This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

\$u \GTSdirectedAdjacencyUpMonochromeRelation/ v\$

or \$u \GTSdirAdjUpMonoRel/ v\$

or \$u \GTSdirAdjUpMonRel/ v\$

or \$u \GTSdAUMR/ v\$

will yield $u \overset{\circ}{\curvearrowright} v$.

\GTSnegatedDirectedAdjacencyUpMonochrome, \GTSnegDirAdjUpMono, \GTSnegDirAdjUpMon, \GTSnDAUM This macro will draw a monochrome negated directed adjacency upward symbol.

A.\GTSnegatedDirectedAdjacencyUpMonochrome/A.

or A.\GTSnegDirAdjUpMono/A.

or A.\GTSnegDirAdjUpMon/A.

or A.\GTSnDAUM/A.

will yield $A \overset{\circ}{\curvearrowleft} A$.

u \GTSnegatedDirectedAdjacencyUpMonochrome/ v

or u \GTSnegDirAdjUpMono/ v

or u \GTSnegDirAdjUpMon/ v

or u \GTSnDAUM/ v

will yield $u \overset{\circ}{\curvearrowleft} v$.

\$u \GTSnegatedDirectedAdjacencyUpMonochrome/ v\$

or \$u \GTSnegDirAdjUpMono/ v\$

or \$u \GTSnegDirAdjUpMon/ v\$

or \$u \GTSnDAUM/ v\$

will yield $u \overset{\circ}{\curvearrowleft} v$.

\GTSnegatedDirectedAdjacencyUpMonochromeRelation, \GTSnegDirAdjUpMonoRel, \GTSnegDirAdjUpMonRel, \GTSnDAUMR This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

\$u \GTSnegatedDirectedAdjacencyUpMonochromeRelation/ v\$

or \$u \GTSnegDirAdjUpMonoRel/ v\$

or \$u \GTSnegDirAdjUpMonRel/ v\$

or \$u \GTSnDAUMR/ v\$

will yield $u \overset{\circ}{\curvearrowleft} v$.

\GTSdirectedAdjacencyDownMonochrome, \GTSdirAdjDownMono, \GTSdirAdjDownMon, \GTSdADM This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

$A.\text{\textbackslash GTSdirectedAdjacencyDownMonochrome}/A.$
or $A.\text{\textbackslash GTSdirAdjDownMono}/A.$
or $A.\text{\textbackslash GTSdirAdjDownMon}/A.$
or $A.\text{\textbackslash GTSdADM}/A.$
will yield $A.\text{\textcircled{\tiny A}}.$
 $u\text{\textbackslash GTSdirectedAdjacencyDownMonochrome}/v$
or $u\text{\textbackslash GTSdirAdjDownMono}/v$
or $u\text{\textbackslash GTSdirAdjDownMon}/v$
or $u\text{\textbackslash GTSdADM}/v$
will yield $u\text{\textcircled{\tiny v}}.$
 $\$u\text{\textbackslash GTSdirectedAdjacencyDownMonochrome}/v\$$
or $\$u\text{\textbackslash GTSdirAdjDownMono}/v\$$
or $\$u\text{\textbackslash GTSdirAdjDownMon}/v\$$
or $\$u\text{\textbackslash GTSdADM}/v\$$
will yield $u\text{\textcircled{\tiny v}}.$

$\text{\textbackslash GTSdirectedAdjacencyDownMonochromeRelation}, \text{\textbackslash GTSdirAdjDownMonoRel}, \text{\textbackslash GTSdirAdjDownMonRel}, \text{\textbackslash GTSdADMR}$ This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

$\$u\text{\textbackslash GTSdirectedAdjacencyDownMonochromeRelation}/v\$$
or $\$u\text{\textbackslash GTSdirAdjDownMonoRel}/v\$$
or $\$u\text{\textbackslash GTSdirAdjDownMonRel}/v\$$
or $\$u\text{\textbackslash GTSdADMR}/v\$$
will yield $u\text{\textcircled{\tiny v}}.$

$\text{\textbackslash GTSnegatedDirectedAdjacencyDownMonochrome}, \text{\textbackslash GTSnegDirAdjDownMono}, \text{\textbackslash GTSnegDirAdjDownMon}, \text{\textbackslash GTSnDADM}$ This macro will draw a monochrome negated directed adjacency downward symbol.

$A.\text{\textbackslash GTSnegatedDirectedAdjacencyDownMonochrome}/A.$
or $A.\text{\textbackslash GTSnegDirAdjDownMono}/A.$
or $A.\text{\textbackslash GTSnegDirAdjDownMon}/A.$
or $A.\text{\textbackslash GTSnDADM}/A.$
will yield $A.\text{\textcircled{\tiny A}}.$
 $u\text{\textbackslash GTSnegatedDirectedAdjacencyDownMonochrome}/v$
or $u\text{\textbackslash GTSnegDirAdjDownMono}/v$
or $u\text{\textbackslash GTSnegDirAdjDownMon}/v$
or $u\text{\textbackslash GTSnDADM}/v$
will yield $u\text{\textcircled{\tiny v}}.$
 $\$u\text{\textbackslash GTSnegatedDirectedAdjacencyDownMonochrome}/v\$$
or $\$u\text{\textbackslash GTSnegDirAdjDownMono}/v\$$
or $\$u\text{\textbackslash GTSnegDirAdjDownMon}/v\$$
or $\$u\text{\textbackslash GTSnDADM}/v\$$
will yield $u\text{\textcircled{\tiny v}}.$

$\text{\textbackslash GTSnegatedDirectedAdjacencyDownMonochromeRelation}, \text{\textbackslash GTSnegDirAdjDownMonoRel}, \text{\textbackslash GTSnegDirAdjDownMonRel}, \text{\textbackslash GTSnDADMR}$ This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

$\$u\text{\textbackslash GTSnegatedDirectedAdjacencyDownMonochromeRelation}/v\$$
or $\$u\text{\textbackslash GTSnegDirAdjDownMonoRel}/v\$$
or $\$u\text{\textbackslash GTSnegDirAdjDownMonRel}/v\$$
or $\$u\text{\textbackslash GTSnDADMR}/v\$$
will yield $u\text{\textcircled{\tiny v}}.$

2.2.5 Monochrome incidence macros

$\text{\textbackslash GTSincidenceMonochrome}, \text{\textbackslash GTSincMono}, \text{\textbackslash GTSincMon}, \text{\textbackslash GTSiM}$ This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

$A.\text{\textbackslash GTSincidenceMonochrome}/A.$
or $A.\text{\textbackslash GTSincMono}/A.$
or $A.\text{\textbackslash GTSincMon}/A.$

`or A.\GTSiM/A.`
`will yield  $A \downarrow A$ .`
`v \GTSincidenceMonochrome/ e`
`or v \GTSincMono/ e`
`or v \GTSincMon/ e`
`or v \GTSiM/ e`
`will yield  $v \downarrow e$ .`
`$v \GTSincidenceMonochrome/ e$`
`or $v \GTSincMono/ e$`
`or $v \GTSincMon/ e$`
`or $v \GTSiM/ e$`
`will yield  $v \downarrow e$ .`

`\GTSincidenceMonochromeRelation, \GTSincMonoRel, \GTSincMonRel, \GTSiMR` This macro will draw a monochrome incidence symbol with relation spacing in math mode.

`$v\GTSincidenceMonochromeRelation/e$`
`or $v\GTSincMonoRel/e$`
`or $v\GTSincMonRel/e$`
`or $v\GTSiMR/e$`
`will yield  $v \downarrow e$ .`

`\GTSnegatedIncidenceMonochrome, \GTSnegIncMono, \GTSnegIncMon, \GTSnIM` This macro will draw a monochrome negated incidence symbol.

`A.\GTSnegatedIncidenceMonochrome/A.`
`or A.\GTSnegIncMono/A.`
`or A.\GTSnegIncMon/A.`
`or A.\GTSnIM/A.`
`will yield  $A \downarrow A$ .`
`v \GTSnegatedIncidenceMonochrome/ e`
`or v \GTSnegIncMono/ e`
`or v \GTSnegIncMon/ e`
`or v \GTSnIM/ e`
`will yield  $v \downarrow e$ .`
`$v \GTSnegatedIncidenceMonochrome/ e$`
`or $v \GTSnegIncMono/ e$`
`or $v \GTSnegIncMon/ e$`
`or $v \GTSnIM/ e$`
`will yield  $v \downarrow e$ .`

`\GTSnegatedIncidenceMonochromeRelation, \GTSnegIncMonoRel, \GTSnegIncMonRel, \GTSnIMR` This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

`$v\GTSnegatedIncidenceMonochromeRelation/e$`
`or $v\GTSnegIncMonoRel/e$`
`or $v\GTSnegIncMonRel/e$`
`or $v\GTSnIMR/e$`
`will yield  $v \downarrow e$ .`

`\GTSoutIncidenceMonochrome, \GTSoutIncMono, \GTSoutIncMon, \GTSoIM` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

`A.\GTSoutIncidenceMonochrome/A.`
`or A.\GTSoutIncMono/A.`
`or A.\GTSoutIncMon/A.`
`or A.\GTSoIM/A.`
`will yield  $A \uparrow A$ .`
`v \GTSoutIncidenceMonochrome/ e`
`or v \GTSoutIncMono/ e`
`or v \GTSoutIncMon/ e`
`or v \GTSoIM/ e`
`will yield  $v \uparrow e$ .`

$\$v \backslash \text{GTSoutIncidenceMonochrome/} e\$$
 or $\$v \backslash \text{GTSoutIncMono/} e\$$
 or $\$v \backslash \text{GTSoutIncMon/} e\$$
 or $\$v \backslash \text{GTSOIM/} e\$$
 will yield $v \overleftarrow{e}$.

$\backslash \text{GTSoutIncidenceMonochromeRelation}$, $\backslash \text{GTSoutIncMonoRel}$, $\backslash \text{GTSoutIncMonRel}$, $\backslash \text{GTSOIMR}$
 This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.
 $\$v \backslash \text{GTSoutIncidenceMonochromeRelation/} e\$$
 or $\$v \backslash \text{GTSoutIncMonoRel/} e\$$
 or $\$v \backslash \text{GTSoutIncMonRel/} e\$$
 or $\$v \backslash \text{GTSOIMR/} e\$$
 will yield $v \overrightarrow{e}$.

$\backslash \text{GTSnegatedOutIncidenceMonochrome}$, $\backslash \text{GTSnegOutIncMono}$, $\backslash \text{GTSnegOutIncMon}$, $\backslash \text{GTSnOIM}$
 This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.
 $A.\backslash \text{GTSnegatedOutIncidenceMonochrome/} A.$
 or $A.\backslash \text{GTSnegOutIncMono/} A.$
 or $A.\backslash \text{GTSnegOutIncMon/} A.$
 or $A.\backslash \text{GTSnOIM/} A.$
 will yield $A.\overleftarrow{A}$.
 $v \backslash \text{GTSnegatedOutIncidenceMonochrome/} e$
 or $v \backslash \text{GTSnegOutIncMono/} e$
 or $v \backslash \text{GTSnegOutIncMon/} e$
 or $v \backslash \text{GTSnOIM/} e$
 will yield $v \overleftarrow{e}$.
 $\$v \backslash \text{GTSnegatedOutIncidenceMonochrome/} e\$$
 or $\$v \backslash \text{GTSnegOutIncMono/} e\$$
 or $\$v \backslash \text{GTSnegOutIncMon/} e\$$
 or $\$v \backslash \text{GTSnOIM/} e\$$
 will yield $v \overleftarrow{e}$.

$\backslash \text{GTSnegatedOutIncidenceMonochromeRelation}$, $\backslash \text{GTSnegOutIncMonoRel}$, $\backslash \text{GTSnegOutIncMonRel}$, $\backslash \text{GTSnOIMR}$ This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.
 $\$v \backslash \text{GTSnegatedOutIncidenceMonochromeRelation/} e\$$
 or $\$v \backslash \text{GTSnegOutIncMonoRel/} e\$$
 or $\$v \backslash \text{GTSnegOutIncMonRel/} e\$$
 or $\$v \backslash \text{GTSnOIMR/} e\$$
 will yield $v \overrightarrow{e}$.

$\backslash \text{GTSinIncidenceMonochrome}$, $\backslash \text{GTSinIncMono}$, $\backslash \text{GTSinIncMon}$, $\backslash \text{GTSiIM}$ This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.
 $A.\backslash \text{GTSinIncidenceMonochrome/} A.$
 or $A.\backslash \text{GTSinIncMono/} A.$
 or $A.\backslash \text{GTSinIncMon/} A.$
 or $A.\backslash \text{GTSiIM/} A.$
 will yield $A.\overrightarrow{A}$.
 $v \backslash \text{GTSinIncidenceMonochrome/} e$
 or $v \backslash \text{GTSinIncMono/} e$
 or $v \backslash \text{GTSinIncMon/} e$
 or $v \backslash \text{GTSiIM/} e$
 will yield $v \overrightarrow{e}$.
 $\$v \backslash \text{GTSinIncidenceMonochrome/} e\$$
 or $\$v \backslash \text{GTSinIncMono/} e\$$
 or $\$v \backslash \text{GTSinIncMon/} e\$$
 or $\$v \backslash \text{GTSiIM/} e\$$
 will yield $v \overrightarrow{e}$.

\GTSinIncidenceMonochromeRelation, \GTSinIncMonoRel, \GTSinIncMonRel, \GTSiIMR This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSinIncidenceMonochromeRelation}/e\$$
or $\$v\backslash\text{GTSinIncMonoRel}/e\$$
or $\$v\backslash\text{GTSinIncMonRel}/e\$$
or $\$v\backslash\text{GTSiIMR}/e\$$
will yield $v \downarrow e$.

\GTSnegatedInIncidenceMonochrome, \GTSnegInIncMono, \GTSnegInIncMon, \GTSniIM This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

$A.\backslash\text{GTSnegatedInIncidenceMonochrome}/A.$
or $A.\backslash\text{GTSnegInIncMono}/A.$
or $A.\backslash\text{GTSnegInIncMon}/A.$
or $A.\backslash\text{GTSniIM}/A.$
will yield $A \nmid A$.
 $v \backslash\text{GTSnegatedInIncidenceMonochrome}/ e$
or $v \backslash\text{GTSnegInIncMono}/ e$
or $v \backslash\text{GTSnegInIncMon}/ e$
or $v \backslash\text{GTSniIM}/ e$
will yield $v \nmid e$.
 $\$v \backslash\text{GTSnegatedInIncidenceMonochrome}/ e\$$
or $\$v \backslash\text{GTSnegInIncMono}/ e\$$
or $\$v \backslash\text{GTSnegInIncMon}/ e\$$
or $\$v \backslash\text{GTSniIM}/ e\$$
will yield $v \nmid e$.

\GTSnegatedInIncidenceMonochromeRelation, \GTSnegInIncMonoRel, \GTSnegInIncMonRel, \GTSniIMR This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

$\$v\backslash\text{GTSnegatedInIncidenceMonochromeRelation}/e\$$
or $\$v\backslash\text{GTSnegInIncMonoRel}/e\$$
or $\$v\backslash\text{GTSnegInIncMonRel}/e\$$
or $\$v\backslash\text{GTSniIMR}/e\$$
will yield $v \nmid e$.

2.2.6 Meta-logic macros

\GTSmetaNot, \GTSmN This macro will draw a meta-not or relation-not symbol.

$A.\backslash\text{GTSmetaNot}/A.$
or $A.\backslash\text{GTSmN}/A.$
will yield $A \nmid A$.
 $u \backslash\text{GTSmetaNot}/\backslash\text{GTSdAU}/ v$
or $u \backslash\text{GTSmN}/\backslash\text{GTSdAU}/ v$
will yield $u \nmid_{\text{d}} v$.
 $\$u \backslash\text{GTSmetaNot}/ \backslash\text{GTSdAU}/ v\$$
or $\$u \backslash\text{GTSmN}/ \backslash\text{GTSdAU}/ v\$$
will yield $u \nmid_{\text{d}} v$.

\GTSmetaAnd, \GTSmA This macro will draw a meta-and or relations-and symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

Look at the discussion after all logical symbols for less trivial uses and recommendations.

$A.\backslash\text{GTSmetaAnd}/A.$
or $A.\backslash\text{GTSmA}/A.$
will yield $A \wedge A$.
 $u \backslash\text{GTSdAD}/\backslash\text{GTSmetaAnd}/\backslash\text{GTSdAU}/ v$
or $u \backslash\text{GTSdAD}/\backslash\text{GTSmA}/\backslash\text{GTSdAU}/ v$
will yield $u \wedge_{\text{d}} v$.

`$u \GTSdAD/ \GTSmetaAnd/ \GTSdAU/ v$`
or `$u \GTSdAD/ \GTSmA/ \GTSdAU/ v$`
will yield $u \overset{\text{m}}{\underset{\text{a}}{\wedge}} v$.

`\GTSmetaAndRelation, \GTSmetaAndRel, \GTSmAR` This macro will draw a meta-and symbol with relation spacing in math mode.

`$u \GTSdAD/\GTSmetaAndRelation/\GTSdAU/ v$`
or `$u \GTSdAD/\GTSmetaAndRel/\GTSdAU/ v$`
or `$u \GTSdAD/\GTSmAR/\GTSdAU/ v$`
will yield $u \overset{\text{r}}{\underset{\text{a}}{\wedge}} v$.

`\GTSmetaOr, \GTSmO` This macro will draw a meta-or or relations-or symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

Look at the discussion after all logical symbols for less trivial uses and recommendations.

`A.\GTSmetaOr/A.`

or `A.\GTSmO/A.`

will yield $A \overset{\text{r}}{\vee} A$.

`u \GTSdAD/\GTSmetaOr/\GTSdAU/ v`

or `u \GTSdAD/\GTSmO/\GTSdAU/ v`

will yield $u \overset{\text{r}}{\underset{\text{O}}{\vee}} v$.

`$u \GTSdAD/ \GTSmetaOr/ \GTSdAU/ v$`

or `$u \GTSdAD/ \GTSmO/ \GTSdAU/ v$`

will yield $u \overset{\text{r}}{\underset{\text{O}}{\vee}} v$.

`\GTSmetaOrRelation, \GTSmetaOrRel, \GTSmOR` This macro will draw a meta-or symbol with relation spacing in math mode.

`$u \GTSdAD/\GTSmetaOrRelation/\GTSdAU/ v$`

or `$u \GTSdAD/\GTSmetaOrRel/\GTSdAU/ v$`

or `$u \GTSdAD/\GTSmOR/\GTSdAU/ v$`

will yield $u \overset{\text{r}}{\underset{\text{O}}{\vee}} v$.

`\GTSxmetallogic, \GTSxml` Versatile macros for meta-logic symbols that can be given options.

Argument:

`[\langle optionsList \rangle]` Sets the following options if not empty:

`not` or `n`,

`and` or `a`,

`or` or `o`,

`relation` or `rel` or `r`.

Of course first three options are incompatible, and the fourth is compatible only with the second and third options.

`\GTSx` One macro to rule them all. Versatile macro for all symbols that can be given options, and a symbol type mandatory switch.

Argument:

`[\langle optionsList \rangle]` Sets the following options if not empty:

- If $\{\langle symbolType \rangle\}$ below is `a`, see `\GTSxa`:
`mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,
`secondColor=color2` or `secondC=color2` or `sColor=color2` or `sC=color2`,
`monochrome` or `mono` or `mon` or `m` (equivalent to `mC=black,sC=black`),
`negated` or `not` or `n`,
`up` or `u`,
`down` or `d`,
`relation` or `rel` or `r`.
- If $\{\langle symbolType \rangle\}$ below is `i`, see `\GTSxi`:
`mainColor=color1` or `mainC=color1` or `mColor=color1` or `mC=color1`,

- If $\{\langle symbolType \rangle\}$ below is m, see \GTSxml:
not or n,
and or a,
or or o,
relation or rel or r.

2.3 Good practices

$$\forall u, v, u \left[\begin{array}{c} \textcircled{\text{u}} \textcircled{\text{u}} \textcircled{\text{u}} \\ \textcircled{\text{u}} \textcircled{\text{u}} \textcircled{\text{u}} \end{array} \right] v.$$
$$\exists v, e, v \left[\overset{m}{V} \begin{array}{c} \text{---} \text{---} \text{---} \end{array} \right] e.$$

3 Implementation

18

```

16 \def\GTS@drawDownArrow{}
17
\GTS@newcommand This macro will enforce that the other macros are called with a trailing slash, avoiding
problems with spaces after macro call.
Arguments:
{\commandname} The name of the command to create.
{\commandcode} The code of the command to create.
18 \ifdefinable{\GTS@newcommand}\def\GTS@newcommand#1#2{%
19   \ifdefinable{#1}\def#1/{#2}}
20 }}

\GTS@newprotectedcommand This macro will enforce that the other macros are called with a trailing
slash, avoiding problems with spaces after macro call.
Arguments:
{\commandname} The name of the command to create.
{\commandcode} The code of the command to create.
21 \ifdefinable{\GTS@newprotectedcommand}\def\GTS@newprotectedcommand#1#2{%
22   \ifdefinable{#1}\protected\def#1/{#2}}
23 }}

\GTSsetMainColor This macro will set the main color used in GraphTheorySymbol. This color has initial
value “black”.
Argument:
{\color} The color that must be used by GraphTheorySymbol afterward.
24 \newcommand\GTSsetMainColor[1]{%
25   \edef\GTS@mainColorTemp{#1}%
26   \edef\GTS@mainColorBackup{\GTS@mainColor}%
27   \edef\GTS@mainColor{\GTS@mainColorTemp}%
28 }

\GTSsetSecondColor This macro will set the second color used in GraphTheorySymbol. This color has
initial value “gray”.
Argument:
{\color} The color that must be used by GraphTheorySymbol afterward.
29 \newcommand\GTSsetSecondColor[1]{%
30   \edef\GTS@secondColorTemp{#1}%
31   \edef\GTS@secondColorBackup{\GTS@secondColor}%
32   \edef\GTS@secondColor{\GTS@secondColorTemp}%
33 }

\GTS@a This macro will draw any adjacency symbol given the correct parameters. Its use is purely technical.
See the macros derived from it and \GTS@b: \GTSxadjacency, \GTSxadj, and \GTSxa.
Arguments:
{\mainColor} If non-empty, this color will be used as main color.
{\secondColor} If non-empty, this color will be used as second color.
{\drawSlash} Non-empty implies that a negation will be drawn.
{\drawUpArrow} Non-empty implies that an up-arrow will be drawn.
{\drawDownArrow} Non-empty implies that a down-arrow will be drawn.
34 \newcommand\GTS@a[5]{%
35   \if#1\empty\else%
36     \GTSsetMainColor{#1}%
37   \fi%
38   \if#2\empty\else%
39     \GTSsetSecondColor{#2}%
40   \fi%
41   \text{\ensuremath{%
42     \raisebox{-0.32ex}{\scalebox{1.5}[1.5]{%
43       %
44       \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
45         $\mathcolor{\GTS@secondColor}{\circ}$%
46       }}%
47       \hspace{-0.39ex}%
48       \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%

```

```

49 $\mathcolor{\GTS@secondColor}{\bullet}$%
50 }}%
51 \hspace{-0.12ex}%
52 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
53 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
54 }}%
55 \hspace{-0.40ex}%
56 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
57 $\mathcolor{\GTS@secondColor}{\circ}$%
58 }}%
59 \hspace{-0.39ex}%
60 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
61 $\mathcolor{\GTS@secondColor}{\bullet}$%
62 }}%
63 \hspace{0.27ex}%
if GTS@drawUpArrow empty else
64 \if#4\empty\else%
65 \hspace{-0.65ex}%
66 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
67 \textcolor{\GTS@mainColor}{\textasciicircum}%
68 }}%
69 \fi%
if GTS@drawDownArrow empty else
70 \if#5\empty\else%
71 \hspace{-0.65ex}%
72 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
73 \textcolor{\GTS@mainColor}{\textasciicircum}%
74 }}%
75 \fi%
if GTS@drawSlash empty else
76 \if#3\empty\else%
77 \hspace{-0.65ex}%
78 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
79 \textcolor{black}{/}%
80 }}%
81 \fi%
82 %
83 }}%
84 }}%
85 \if#1\empty\else%
86 \GTSsetMainColor{\GTS@mainColorBackup}%
87 \fi%
88 \if#2\empty\else%
89 \GTSsetSecondColor{\GTS@secondColorBackup}%
90 \fi%
91 }

```

\GTS@b This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and \GTS@a: \GTSxadjacency, \GTSxadj, and \GTSxa.

Arguments:

- {\mainColor)} If non-empty, this color will be used as main color.
- {\secondColor)} If non-empty, this color will be used as second color.
- {\drawSlash)} Non-empty implies that a negation will be drawn.
- {\drawUpArrow)} Non-empty implies that an up-arrow will be drawn.
- {\drawDownArrow)} Non-empty implies that a down-arrow will be drawn.
- {\relation)} Non-empty implies that relation spacing will be added.

```

92 \newcommand{\GTS@b}[6]{%
93 \if#61%
94 \mathrel{\GTS@a{#1}{#2}{#3}{#4}{#5}}%
95 \else%
96 \GTS@a{#1}{#2}{#3}{#4}{#5}%
97 \fi%
98 }

```

\GTSxadjacency, \GTSxadj, \GTSxa Versatile macros for adjacency that can be given options.

Argument:

[*\optionsList*] Sets the following options if not empty:

mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
up or u,
down or d,
relation or rel or r.

```

99 \define@key{GTSxa}{mainColor}{\def\GTS@@mainColor{#1}}
100 \define@key{GTSxa}{mainC}{\def\GTS@@mainColor{#1}}
101 \define@key{GTSxa}{mColor}{\def\GTS@@mainColor{#1}}
102 \define@key{GTSxa}{mC}{\def\GTS@@mainColor{#1}}
103 \define@key{GTSxa}{secondColor}{\def\GTS@@secondColor{#1}}
104 \define@key{GTSxa}{secondC}{\def\GTS@@secondColor{#1}}
105 \define@key{GTSxa}{sColor}{\def\GTS@@secondColor{#1}}
106 \define@key{GTSxa}{sC}{\def\GTS@@secondColor{#1}}
107 \define@key{GTSxa}{monochrome}[1]%
108 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
109 \define@key{GTSxa}{mono}[1]%
110 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
111 \define@key{GTSxa}{mon}[1]%
112 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
113 \define@key{GTSxa}{m}[1]%
114 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
115 \define@key{GTSxa}{negated}[1]{\def\GTS@@drawSlash{1}}
116 \define@key{GTSxa}{not}[1]{\def\GTS@@drawSlash{1}}
117 \define@key{GTSxa}{n}[1]{\def\GTS@@drawSlash{1}}
118 \define@key{GTSxa}{up}[1]{\def\GTS@@drawUpArrow{1}}
119 \define@key{GTSxa}{u}[1]{\def\GTS@@drawUpArrow{1}}
120 \define@key{GTSxa}{down}[1]{\def\GTS@@drawDownArrow{1}}
121 \define@key{GTSxa}{d}[1]{\def\GTS@@drawDownArrow{1}}
122 \define@key{GTSxa}{relation}[1]{\def\GTS@@relation{1}}
123 \define@key{GTSxa}{rel}[1]{\def\GTS@@relation{1}}
124 \define@key{GTSxa}{r}[1]{\def\GTS@@relation{1}}
125
126 \newcommand{\GTSxa}[1] [] {%
127 \def\GTS@@mainColor{}%
128 \def\GTS@@secondColor{}%
129 \def\GTS@@drawSlash{}%
130 \def\GTS@@drawUpArrow{}%
131 \def\GTS@@drawDownArrow{}%
132 \def\GTS@@relation{}%
133 \setkeys{GTSxa}{#1}%
134 \GTS@b{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
135 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
136 }
137
138 \newcommand{\GTSxadj}[1] [] {\GTSxa[#1]}
139
140 \newcommand{\GTSxadjacency}[1] [] {\GTSxa[#1]}

```

\GTSadjacency, \GTSadj, \GTSa This macro will draw an adjacency symbol.

```

141 \GTS@newcommand{\GTSadjacency}{\GTS@a{}{}{}{}}
142 \GTS@newcommand{\GTSadj}{\GTS@a{}{}{}{}}
143 \GTS@newcommand{\GTSa}{\GTS@a{}{}{}{}}

```

\GTSadjacencyRelation, \GTSadjRel, \GTSaR This macro will draw an adjacency symbol with relation spacing in math mode.

```

144 \GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSa/}}
145 \GTS@newcommand{\GTSadjRel}{\mathrel{\GTSa/}}
146 \GTS@newcommand{\GTSaR}{\mathrel{\GTSa/}}

```

\GTSunadjacency, \GTSunadj, \GTSu This macro will draw an unadjacency symbol.

```

147 \GTS@newcommand{\GTSunadjacency}{\GTS@a{}{}{1}{}{}}
148 \GTS@newcommand{\GTSunadj}{\GTS@a{}{}{1}{}{}}
149 \GTS@newcommand{\GTSu}{\GTS@a{}{}{1}{}{}}

```

\GTSunadjacencyRelation, \GTSunadjRel, \GTSuR This macro will draw an unadjacency symbol with relation spacing in math mode.

```

150 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSu/}}
151 \GTS@newcommand{\GTSunadjRel}{\mathrel{\GTSu/}}
152 \GTS@newcommand{\GTSuR}{\mathrel{\GTSu/}}

```

\GTSdirectedAdjacencyUp, \GTSdirAdjUp, \GTSdAU This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```

153 \GTS@newcommand{\GTSdirectedAdjacencyUp}{\GTS@a{}{}{1}{}{}}
154 \GTS@newcommand{\GTSdirAdjUp}{\GTS@a{}{}{1}{}{}}
155 \GTS@newcommand{\GTSdAU}{\GTS@a{}{}{1}{}{}}

```

\GTSdirectedAdjacencyUpRelation, \GTSdirAdjUpRel, \GTSdAUR This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

```

156 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{\mathrel{\GTSdAU/}}
157 \GTS@newcommand{\GTSdirAdjUpRel}{\mathrel{\GTSdAU/}}
158 \GTS@newcommand{\GTSdAUR}{\mathrel{\GTSdAU/}}

```

\GTSnegatedDirectedAdjacencyUp, \GTSnegDirAdjUp, \GTSnDAU This macro will draw a negated directed adjacency upward symbol.

```

159 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{\GTS@a{}{}{1}{1}{}{}}
160 \GTS@newcommand{\GTSnegDirAdjUp}{\GTS@a{}{}{1}{1}{}{}}
161 \GTS@newcommand{\GTSnDAU}{\GTS@a{}{}{1}{1}{}{}}

```

\GTSnegatedDirectedAdjacencyUpRelation, \GTSnegDirAdjUpRel, \GTSnDAUR This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

```

162 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
163 \mathrel{\GTSnDAU/}%
164 }
165 \GTS@newcommand{\GTSnegDirAdjUpRel}{\mathrel{\GTSnDAU/}}
166 \GTS@newcommand{\GTSnDAUR}{\mathrel{\GTSnDAU/}}

```

\GTSdirectedAdjacencyDown, \GTSdirAdjDown, \GTSdAD This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```

167 \GTS@newcommand{\GTSdirectedAdjacencyDown}{\GTS@a{}{}{}{1}{}{}}
168 \GTS@newcommand{\GTSdirAdjDown}{\GTS@a{}{}{}{1}{}{}}
169 \GTS@newcommand{\GTSdAD}{\GTS@a{}{}{}{1}{}{}}

```

\GTSdirectedAdjacencyDownRelation, \GTSdirAdjDownRel, \GTSdADR This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

```

170 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{\mathrel{\GTSdAD/}}
171 \GTS@newcommand{\GTSdirAdjDownRel}{\mathrel{\GTSdAD/}}
172 \GTS@newcommand{\GTSdADR}{\mathrel{\GTSdAD/}}

```

\GTSnegatedDirectedAdjacencyDown, \GTSnegDirAdjDown, \GTSnDAD This macro will draw a negated directed adjacency downward symbol.

```

173 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{\GTS@a{}{}{1}{}{1}{}{}}
174 \GTS@newcommand{\GTSnegDirAdjDown}{\GTS@a{}{}{1}{}{1}{}{}}
175 \GTS@newcommand{\GTSnDAD}{\GTS@a{}{}{1}{}{1}{}{}}

```

\GTSnegatedDirectedAdjacencyDownRelation, \GTSnegDirAdjDownRel, \GTSnDADR This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

```

176 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
177 \mathrel{\GTSnDAD/}%
178 }
179 \GTS@newcommand{\GTSnegDirAdjDownRel}{\mathrel{\GTSnDAD/}}
180 \GTS@newcommand{\GTSnDADR}{\mathrel{\GTSnDAD/}}

```

\GTS@i This macro will draw any incidence symbol given the correct parameters. Its use is purely technical. See the macros derived from it and **\GTS@j**: **\GTSxincidence**, **\GTSxinc**, and **\GTSxi**. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

Arguments:

- {\mainColor}** If non-empty, this color will be used as main color.
- {\secondColor}** If non-empty, this color will be used as second color.
- {\drawSlash}** Non-empty implies that a negation will be drawn.
- {\drawUpArrow}** Non-empty implies that an up-arrow will be drawn.
- {\drawDownArrow}** Non-empty implies that a down-arrow will be drawn.

```

181 \newcommand{\GTS@i}[5]{%
182 \if#1\empty\else%
183 \GTSsetMainColor{#1}%
184 \fi%
185 \if#2\empty\else%
186 \GTSsetSecondColor{#2}%
187 \fi%
188 \text{\ensuremath{%
189 \raisebox{-0.22ex}{\scalebox{2.05}[2.05]{%
190 %
191 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
192 $\mathcolor{\GTS@secondColor}{\circ}$%
193 }}%
194 \hspace{-0.39ex}%
195 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
196 $\mathcolor{\GTS@secondColor}{\bullet}$%
197 }}%
198 \hspace{-0.12ex}%
199 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
200 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
201 }}%
202 \hspace{-0.18ex}%
203 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
204 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
205 }}%
206 \hspace{0.2ex}%
if GTS@drawUpArrow empty else
207 \if#4\empty\else%
208 \hspace{-0.65ex}%
209 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
210 \textcolor{\GTS@secondColor}{\textasciicircum}%
211 }}%
212 \fi%
if GTS@drawDownArrow empty else
213 \if#5\empty\else%
214 \hspace{-0.65ex}%
215 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
216 \textcolor{\GTS@secondColor}{\textasciicircum}%
217 }}%
218 \fi%
if GTS@drawSlash empty else
219 \if#3\empty\else%
220 \hspace{-0.65ex}%
221 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
222 \textcolor{black}{/}%
223 }}%
224 \fi%
225 %
226 }}%
227 }}%
228 \if#1\empty\else%
229 \GTSsetMainColor{\GTS@mainColorBackup}%
230 \fi%
```

```

231 \if#2\empty\else%
232 \GTSsetSecondColor{\GTS@secondColorBackup}%
233 \fi%
234 }

```

\GTS@j This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and \GTS@i: \GTSxincidence, \GTSxinc, and \GTSxi.

Arguments:

{\mainColor} If non-empty, this color will be used as main color.
{\secondColor} If non-empty, this color will be used as second color.
{\drawSlash} Non-empty implies that a negation will be drawn.
{\drawUpArrow} Non-empty implies that an up-arrow will be drawn.
{\drawDownArrow} Non-empty implies that a down-arrow will be drawn.
{\relation} Non-empty implies that relation spacing will be added.

```

235 \newcommand{\GTS@j}[6]{%
236 \if#61%
237 \mathrel{\GTS@i{#1}{#2}{#3}{#4}{#5}}%
238 \else%
239 \GTS@i{#1}{#2}{#3}{#4}{#5}%
240 \fi%
241 }

```

\GTSxincidence, \GTSxinc, \GTSxi Versatile macros for incidence that can be given options.

Argument:

[{\optionsList}] Sets the following options if not empty:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
in or i,
out or o,
relation or rel or r.

```

242 \define@key{GTSxi}{mainColor}{\def\GTS@@mainColor{#1}}
243 \define@key{GTSxi}{mainC}{\def\GTS@@mainColor{#1}}
244 \define@key{GTSxi}{mColor}{\def\GTS@@mainColor{#1}}
245 \define@key{GTSxi}{mC}{\def\GTS@@mainColor{#1}}
246 \define@key{GTSxi}{secondColor}{\def\GTS@@secondColor{#1}}
247 \define@key{GTSxi}{secondC}{\def\GTS@@secondColor{#1}}
248 \define@key{GTSxi}{sColor}{\def\GTS@@secondColor{#1}}
249 \define@key{GTSxi}{sC}{\def\GTS@@secondColor{#1}}
250 \define@key{GTSxi}{monochrome}[1]%
251 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
252 \define@key{GTSxi}{mono}[1]%
253 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
254 \define@key{GTSxi}{mon}[1]%
255 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
256 \define@key{GTSxi}{m}[1]%
257 {\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
258 \define@key{GTSxi}{negated}[1]{\def\GTS@@drawSlash{1}}
259 \define@key{GTSxi}{not}[1]{\def\GTS@@drawSlash{1}}
260 \define@key{GTSxi}{n}[1]{\def\GTS@@drawSlash{1}}
261 \define@key{GTSxi}{out}[1]{\def\GTS@@drawUpArrow{1}}
262 \define@key{GTSxi}{o}[1]{\def\GTS@@drawUpArrow{1}}
263 \define@key{GTSxi}{in}[1]{\def\GTS@@drawDownArrow{1}}
264 \define@key{GTSxi}{i}[1]{\def\GTS@@drawDownArrow{1}}
265 \define@key{GTSxi}{relation}[1]{\def\GTS@@relation{1}}
266 \define@key{GTSxi}{rel}[1]{\def\GTS@@relation{1}}
267 \define@key{GTSxi}{r}[1]{\def\GTS@@relation{1}}
268
269 \newcommand{\GTSxi}[1][{}]{%
270 \def\GTS@@mainColor{}%
271 \def\GTS@@secondColor{}%
272 \def\GTS@@drawSlash{}%
273 \def\GTS@@drawUpArrow{}%
274 \def\GTS@@drawDownArrow{}%

```

```

275 \def\GTS@@relation{}%
276 \setkeys{GTSxi}{#1}%
277 \GTS@j{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
278 {\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
279 }
280
281 \newcommand{\GTSxinc}[1][\GTSxi{#1}]
282
283 \newcommand{\GTSxincidence}[1][\GTSxi{#1}]

```

\GTSincidence, \GTSinc, \GTSi This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

284 \GTS@newcommand{\GTSincidence}{\GTS@i{}{}{}{}}
285 \GTS@newcommand{\GTSinc}{\GTS@i{}{}{}{}}
286 \GTS@newcommand{\GTSi}{\GTS@i{}{}{}{}}

```

\GTSincidenceRelation, \GTSincRel, \GTSiR This macro will draw an incidence symbol with relation spacing in math mode.

```

287 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSi/}}
288 \GTS@newcommand{\GTSincRel}{\mathrel{\GTSi/}}
289 \GTS@newcommand{\GTSiR}{\mathrel{\GTSi/}}

```

\GTSnegatedIncidence, \GTSnegInc, \GTSnI This macro will draw a negated incidence symbol.

```

290 \GTS@newcommand{\GTSnegatedIncidence}{\GTS@i{}{}{1}{}{}}
291 \GTS@newcommand{\GTSnegInc}{\GTS@i{}{}{1}{}{}}
292 \GTS@newcommand{\GTSnI}{\GTS@i{}{}{1}{}{}}

```

\GTSnegatedIncidenceRelation, \GTSnegIncRel, \GTSnIR This macro will draw a negated incidence symbol with relation spacing in math mode.

```

293 \GTS@newcommand{\GTSnegatedIncidenceRelation}{\mathrel{\GTSnI/}}
294 \GTS@newcommand{\GTSnegIncRel}{\mathrel{\GTSnI/}}
295 \GTS@newcommand{\GTSnIR}{\mathrel{\GTSnI/}}

```

\GTSoutIncidence, \GTSoutInc, \GTSoI This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

296 \GTS@newcommand{\GTSoutIncidence}{\GTS@i{}{}{1}{}{}}
297 \GTS@newcommand{\GTSoutInc}{\GTS@i{}{}{1}{}{}}
298 \GTS@newcommand{\GTSoI}{\GTS@i{}{}{1}{}{}}

```

\GTSoutIncidenceRelation, \GTSoutIncRel, \GTSoIR This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

299 \GTS@newcommand{\GTSoutIncidenceRelation}{\mathrel{\GTSoI/}}
300 \GTS@newcommand{\GTSoutIncRel}{\mathrel{\GTSoI/}}
301 \GTS@newcommand{\GTSoIR}{\mathrel{\GTSoI/}}

```

\GTSnegatedOutIncidence, \GTSnegOutInc, \GTSnOI This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

```

302 \GTS@newcommand{\GTSnegatedOutIncidence}{\GTS@i{}{}{1}{1}{}{}}
303 \GTS@newcommand{\GTSnegOutInc}{\GTS@i{}{}{1}{1}{}{}}
304 \GTS@newcommand{\GTSnOI}{\GTS@i{}{}{1}{1}{}{}}

```

\GTSnegatedOutIncidenceRelation, \GTSnegOutIncRel, \GTSnOIR This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

305 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{\mathrel{\GTSnOI/}}
306 \GTS@newcommand{\GTSnegOutIncRel}{\mathrel{\GTSnOI/}}
307 \GTS@newcommand{\GTSnOIR}{\mathrel{\GTSnOI/}}

```

\GTSinIncidence, \GTSinInc, \GTSiI This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

308 \GTS@newcommand{\GTSinIncidence}{\GTS@i{}{}{}{1}}
309 \GTS@newcommand{\GTSinInc}{\GTS@i{}{}{}{1}}
310 \GTS@newcommand{\GTSiI}{\GTS@i{}{}{}{1}}

```

\GTSinIncidenceRelation, \GTSinIncRel, \GTSiIR This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
311 \GTS@newcommand{\GTSinIncidenceRelation}{\mathrel{\GTSiI/}}
312 \GTS@newcommand{\GTSinIncRel}{\mathrel{\GTSiI/}}
313 \GTS@newcommand{\GTSiIR}{\mathrel{\GTSiI/}}
```

\GTSnegatedInIncidence, \GTSnegInInc, \GTSnII This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

```
314 \GTS@newcommand{\GTSnegatedInIncidence}{\GTS@i{}{}{1}{}{1}}
315 \GTS@newcommand{\GTSnegInInc}{\GTS@i{}{}{1}{}{1}}
316 \GTS@newcommand{\GTSnII}{\GTS@i{}{}{1}{}{1}}
```

\GTSnegatedInIncidenceRelation, \GTSnegInIncRel, \GTSnIIR This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
317 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{\mathrel{\GTSnII/}}
318 \GTS@newcommand{\GTSnegInIncRel}{\mathrel{\GTSnII/}}
319 \GTS@newcommand{\GTSnIIR}{\mathrel{\GTSnII/}}
```

\GTSadjacencyMonochrome, \GTSadjMono, \GTSadjMon, \GTSaM This macro will draw a monochrome adjacency symbol.

```
320 \GTS@newcommand{\GTSadjacencyMonochrome}{\GTS@a{black}{black}{}{}{}}
321 \GTS@newcommand{\GTSadjMono}{\GTS@a{black}{black}{}{}{}}
322 \GTS@newcommand{\GTSadjMon}{\GTS@a{black}{black}{}{}{}}
323 \GTS@newcommand{\GTSaM}{\GTS@a{black}{black}{}{}{}}
```

\GTSadjacencyMonochromeRelation, \GTSadjMonoRel, \GTSadjMonRel, \GTSaMR This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

```
324 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{\mathrel{\GTSaM/}}
325 \GTS@newcommand{\GTSadjMonoRel}{\mathrel{\GTSaM/}}
326 \GTS@newcommand{\GTSadjMonRel}{\mathrel{\GTSaM/}}
327 \GTS@newcommand{\GTSaMR}{\mathrel{\GTSaM/}}
```

\GTSunadjacencyMonochrome, \GTSunadjMono, \GTSunadjMon, \GTSuM This macro will draw a monochrome unadjacency symbol.

```
328 \GTS@newcommand{\GTSunadjacencyMonochrome}{\GTS@a{black}{black}{1}{}{}}
329 \GTS@newcommand{\GTSunadjMono}{\GTS@a{black}{black}{1}{}{}}
330 \GTS@newcommand{\GTSunadjMon}{\GTS@a{black}{black}{1}{}{}}
331 \GTS@newcommand{\GTSuM}{\GTS@a{black}{black}{1}{}{}}
```

\GTSunadjacencyMonochromeRelation, \GTSunadjMonoRel, \GTSunadjMonRel, \GTSuMR This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

```
332 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{\mathrel{\GTSuM/}}
333 \GTS@newcommand{\GTSunadjMonoRel}{\mathrel{\GTSuM/}}
334 \GTS@newcommand{\GTSunadjMonRel}{\mathrel{\GTSuM/}}
335 \GTS@newcommand{\GTSuMR}{\mathrel{\GTSuM/}}
```

\GTSdirectedAdjacencyUpMonochrome, \GTSdirAdjUpMono, \GTSdirAdjUpMon, \GTSdAUM This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```
336 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
337 \GTS@a{black}{black}{}{1}{}%
338 }
339 \GTS@newcommand{\GTSdirAdjUpMono}{\GTS@a{black}{black}{}{1}{}{}}
340 \GTS@newcommand{\GTSdirAdjUpMon}{\GTS@a{black}{black}{}{1}{}{}}
341 \GTS@newcommand{\GTSdAUM}{\GTS@a{black}{black}{}{1}{}{}}
```

\GTSdirectedAdjacencyUpMonochromeRelation, \GTSdirAdjUpMonoRel, \GTSdirAdjUpMonRel, \GTSdAUMR This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

```
342 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
343 \mathrel{\GTSdAUM/}%
344 }
345 \GTS@newcommand{\GTSdirAdjUpMonoRel}{\mathrel{\GTSdAUM/}}
346 \GTS@newcommand{\GTSdirAdjUpMonRel}{\mathrel{\GTSdAUM/}}
347 \GTS@newcommand{\GTSdAUMR}{\mathrel{\GTSdAUM/}}
```

\GTSnegatedDirectedAdjacencyUpMonochrome, \GTSnegDirAdjUpMono, \GTSnegDirAdjUpMon, \GTSnDAUM This macro will draw a monochrome negated directed adjacency upward symbol.

```
348 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%
349 \GTS@a{black}{black}{1}{1}{}}%
350 }
351 \GTS@newcommand{\GTSnegDirAdjUpMono}{\GTS@a{black}{black}{1}{1}{}}
352 \GTS@newcommand{\GTSnegDirAdjUpMon}{\GTS@a{black}{black}{1}{1}{}}
353 \GTS@newcommand{\GTSnDAUM}{\GTS@a{black}{black}{1}{1}{}}
```

\GTSnegatedDirectedAdjacencyUpMonochromeRelation, \GTSnegDirAdjUpMonoRel, \GTSnegDirAdjUpMonRel, \GTSnDAUMR This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

```
354 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%
355 \mathrel{\GTSnDAUM/}}%
356 }
357 \GTS@newcommand{\GTSnegDirAdjUpMonoRel}{\mathrel{\GTSnDAUM/}}
358 \GTS@newcommand{\GTSnegDirAdjUpMonRel}{\mathrel{\GTSnDAUM/}}
359 \GTS@newcommand{\GTSnDAUMR}{\mathrel{\GTSnDAUM/}}
```

\GTSdirectedAdjacencyDownMonochrome, \GTSdirAdjDownMono, \GTSdirAdjDownMon, \GTSdADM This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```
360 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%
361 \GTS@a{black}{black}{}}{1}}%
362 }
363 \GTS@newcommand{\GTSdirAdjDownMono}{\GTS@a{black}{black}{}}{1}}
364 \GTS@newcommand{\GTSdirAdjDownMon}{\GTS@a{black}{black}{}}{1}}
365 \GTS@newcommand{\GTSdADM}{\GTS@a{black}{black}{}}{1}}
```

\GTSdirectedAdjacencyDownMonochromeRelation, \GTSdirAdjDownMonoRel, \GTSdirAdjDownMonRel, \GTSdADMR This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

```
366 \GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%
367 \mathrel{\GTSdADM/}}%
368 }
369 \GTS@newcommand{\GTSdirAdjDownMonoRel}{\mathrel{\GTSdADM/}}
370 \GTS@newcommand{\GTSdirAdjDownMonRel}{\mathrel{\GTSdADM/}}
371 \GTS@newcommand{\GTSdADMR}{\mathrel{\GTSdADM/}}
```

\GTSnegatedDirectedAdjacencyDownMonochrome, \GTSnegDirAdjDownMono, \GTSnegDirAdjDownMon, \GTSnDADM This macro will draw a monochrome negated directed adjacency downward symbol.

```
372 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%
373 \GTS@a{black}{black}{1}{1}{}}%
374 }
375 \GTS@newcommand{\GTSnegDirAdjDownMono}{\GTS@a{black}{black}{1}{1}{}}
376 \GTS@newcommand{\GTSnegDirAdjDownMon}{\GTS@a{black}{black}{1}{1}{}}
377 \GTS@newcommand{\GTSnDADM}{\GTS@a{black}{black}{1}{1}{}}
```

\GTSnegatedDirectedAdjacencyDownMonochromeRelation, \GTSnegDirAdjDownMonoRel, \GTSnegDirAdjDownMonRel, \GTSnDADMR This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

```
378 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%
379 \mathrel{\GTSnDADM/}}%
380 }
381 \GTS@newcommand{\GTSnegDirAdjDownMonoRel}{\mathrel{\GTSnDADM/}}
382 \GTS@newcommand{\GTSnegDirAdjDownMonRel}{\mathrel{\GTSnDADM/}}
383 \GTS@newcommand{\GTSnDADMR}{\mathrel{\GTSnDADM/}}
```

\GTSincidenceMonochrome, \GTSincMono, \GTSincMon, \GTSiM This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```
384 \GTS@newcommand{\GTSincidenceMonochrome}{\GTS@i{black}{black}{}}{}}
```

```

385 \GTS@newcommand{\GTSincMono}{\GTS@i{black}{black}{-}{-}{-}}
386 \GTS@newcommand{\GTSincMon}{\GTS@i{black}{black}{-}{-}{-}}
387 \GTS@newcommand{\GTSiM}{\GTS@i{black}{black}{-}{-}{-}}

```

\GTSincidenceMonochromeRelation, \GTSincMonoRel, \GTSincMonRel, \GTSiMR This macro will draw a monochrome incidence symbol with relation spacing in math mode.

```

388 \GTS@newcommand{\GTSincidenceMonochromeRelation}{\mathrel{\GTSiM/}}
389 \GTS@newcommand{\GTSincMonoRel}{\mathrel{\GTSiM/}}
390 \GTS@newcommand{\GTSincMonRel}{\mathrel{\GTSiM/}}
391 \GTS@newcommand{\GTSiMR}{\mathrel{\GTSiM/}}

```

\GTSnegatedIncidenceMonochrome, \GTSnegIncMono, \GTSnegIncMon, \GTSnIM This macro will draw a monochrome negated incidence symbol.

```

392 \GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%
393 \GTS@i{black}{black}{1}{-}{-}}
394 }
395 \GTS@newcommand{\GTSnegIncMono}{\GTS@i{black}{black}{1}{-}{-}}
396 \GTS@newcommand{\GTSnegIncMon}{\GTS@i{black}{black}{1}{-}{-}}
397 \GTS@newcommand{\GTSnIM}{\GTS@i{black}{black}{1}{-}{-}}

```

\GTSnegatedIncidenceMonochromeRelation, \GTSnegIncMonoRel, \GTSnegIncMonRel, \GTSnIMR This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

```

398 \GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{\mathrel{\GTSnIM/}}
399 \GTS@newcommand{\GTSnegIncMonoRel}{\mathrel{\GTSnIM/}}
400 \GTS@newcommand{\GTSnegIncMonRel}{\mathrel{\GTSnIM/}}
401 \GTS@newcommand{\GTSnIMR}{\mathrel{\GTSnIM/}}

```

\GTSoutIncidenceMonochrome, \GTSoutIncMono, \GTSoutIncMon, \GTSoIM This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

402 \GTS@newcommand{\GTSoutIncidenceMonochrome}{\GTS@i{black}{black}{-}{1}{-}}
403 \GTS@newcommand{\GTSoutIncMono}{\GTS@i{black}{black}{-}{1}{-}}
404 \GTS@newcommand{\GTSoutIncMon}{\GTS@i{black}{black}{-}{1}{-}}
405 \GTS@newcommand{\GTSoIM}{\GTS@i{black}{black}{-}{1}{-}}

```

\GTSoutIncidenceMonochromeRelation, \GTSoutIncMonoRel, \GTSoutIncMonRel, \GTSoIMR This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

406 \GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{\mathrel{\GTSoIM/}}
407 \GTS@newcommand{\GTSoutIncMonoRel}{\mathrel{\GTSoIM/}}
408 \GTS@newcommand{\GTSoutIncMonRel}{\mathrel{\GTSoIM/}}
409 \GTS@newcommand{\GTSoIMR}{\mathrel{\GTSoIM/}}

```

\GTSnegatedOutIncidenceMonochrome, \GTSnegOutIncMono, \GTSnegOutIncMon, \GTSnOIM This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

```

410 \GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%
411 \GTS@i{black}{black}{1}{1}{-}}
412 }
413 \GTS@newcommand{\GTSnegOutIncMono}{\GTS@i{black}{black}{1}{1}{-}}
414 \GTS@newcommand{\GTSnegOutIncMon}{\GTS@i{black}{black}{1}{1}{-}}
415 \GTS@newcommand{\GTSnOIM}{\GTS@i{black}{black}{1}{1}{-}}

```

\GTSnegatedOutIncidenceMonochromeRelation, \GTSnegOutIncMonoRel, \GTSnegOutIncMonRel, \GTSnOIMR This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

416 \GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%
417 \mathrel{\GTSnOIM/}}
418 }
419 \GTS@newcommand{\GTSnegOutIncMonoRel}{\mathrel{\GTSnOIM/}}
420 \GTS@newcommand{\GTSnegOutIncMonRel}{\mathrel{\GTSnOIM/}}
421 \GTS@newcommand{\GTSnOIMR}{\mathrel{\GTSnOIM/}}

```

`\GTSinIncidenceMonochrome`, `\GTSinIncMono`, `\GTSinIncMon`, `\GTSiIM` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
422 \GTS@newcommand{\GTSinIncidenceMonochrome}{\GTS@i{black}{black}{-}{-}{1}}
423 \GTS@newcommand{\GTSinIncMono}{\GTS@i{black}{black}{-}{-}{1}}
424 \GTS@newcommand{\GTSinIncMon}{\GTS@i{black}{black}{-}{-}{1}}
425 \GTS@newcommand{\GTSiIM}{\GTS@i{black}{black}{-}{-}{1}}
```

`\GTSinIncidenceMonochromeRelation`, `\GTSinIncMonoRel`, `\GTSinIncMonRel`, `\GTSiIMR` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
426 \GTS@newcommand{\GTSinIncidenceMonochromeRelation}{\mathrel{\GTSiIM/}}
427 \GTS@newcommand{\GTSinIncMonoRel}{\mathrel{\GTSiIM/}}
428 \GTS@newcommand{\GTSinIncMonRel}{\mathrel{\GTSiIM/}}
429 \GTS@newcommand{\GTSiIMR}{\mathrel{\GTSiIM/}}
```

`\GTSnegatedInIncidenceMonochrome`, `\GTSnegInIncMono`, `\GTSnegInIncMon`, `\GTSnIIM` This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

```
430 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
431 \GTS@i{black}{black}{1}{-}{1}%
432 }
433 \GTS@newcommand{\GTSnegInIncMono}{\GTS@i{black}{black}{1}{-}{1}}
434 \GTS@newcommand{\GTSnegInIncMon}{\GTS@i{black}{black}{1}{-}{1}}
435 \GTS@newcommand{\GTSnIIM}{\GTS@i{black}{black}{1}{-}{1}}
```

`\GTSnegatedInIncidenceMonochromeRelation`, `\GTSnegInIncMonoRel`, `\GTSnegInIncMonRel`, `\GTSnIIMR` This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
436 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
437 \mathrel{\GTSnIIM/}%
438 }
439 \GTS@newcommand{\GTSnegInIncMonoRel}{\mathrel{\GTSnIIM/}}
440 \GTS@newcommand{\GTSnegInIncMonRel}{\mathrel{\GTSnIIM/}}
441 \GTS@newcommand{\GTSnIIMR}{\mathrel{\GTSnIIM/}}
```

`\GTSmetaNot`, `\GTSmN` This macro will draw a meta-not or relation-not symbol.

```
442 \GTS@newprotectedcommand{\GTSmN}{%
443 \text{\ensuremath{%
444 \raisebox{-0.44ex}{\scalebox{1}[1]{%
445 \raisebox{0ex}{\scalebox{1}[1]{\not$}}}%
446 \hspace{-1.46ex}%
447 \raisebox{1.2ex}{\scalebox{0.7}[0.7]{m}}}%
448 \hspace{0.2ex}%
449 }}}%
450 }}}%
451 }
452 \GTS@newcommand{\GTSmetaNot}{\GTSmN/}
```

`\GTSmetaAnd`, `\GTSmA` This macro will draw a meta-and or relations-and symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

```
453 \GTS@newprotectedcommand{\GTSmA}{%
454 \text{\ensuremath{%
455 \raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
456 \raisebox{0ex}{\scalebox{1}[1]{\wedge$}}}%
457 \hspace{-1.46ex}%
458 \raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}}%
459 }}}%
460 }}}%
461 }
462 \GTS@newcommand{\GTSmetaAnd}{\GTSmA/}
```

\GTSmetaAndRelation, \GTSmetaAndRel, \GTSmAR This macro will draw a meta-and symbol with relation spacing in math mode.

```
463 \GTS@newcommand{\GTSmetaAndRelation}{\mathrel{\GTSmA/}}
464 \GTS@newcommand{\GTSmetaAndRel}{\mathrel{\GTSmA/}}
465 \GTS@newcommand{\GTSmAR}{\mathrel{\GTSmA/}}
```

\GTSmetaOr, \GTSmO This macro will draw a meta-or or relations-or symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures ;). We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublevee or other vee symbols have taken various meanings, whilst we want an unambiguous symbol.

```
466 \GTS@newprotectedcommand{\GTSmO}{%
467 \text{\ensuremath{%
468 \raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
469 \raisebox{0ex}{\scalebox{1}[1]{\vee}}}%
470 \hspace{-1.46ex}%
471 \raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}}%
472 }}%
473 }}%
474 }
475 \GTS@newcommand{\GTSmetaOr}{\GTSmO/}
```

\GTSmetaOrRelation, \GTSmetaOrRel, \GTSmOR This macro will draw a meta-or symbol with relation spacing in math mode.

```
476 \GTS@newcommand{\GTSmetaOrRelation}{\mathrel{\GTSmO/}}
477 \GTS@newcommand{\GTSmetaOrRel}{\mathrel{\GTSmO/}}
478 \GTS@newcommand{\GTSmOR}{\mathrel{\GTSmO/}}
```

\GTSxmetallogic, \GTSxml Versatile macros for meta-logic symbols that can be given options.

Argument:

[*optionsList*] Sets the following options if not empty:

not or n,
and or a,
or or o,
relation or rel or r.

Of course first three options are incompatible, and the fourth is compatible only with the second and third options.

```
479 \define@key{GTSxml}{not}[1]{\def\GTS@@drawMetaNot{1}}
480 \define@key{GTSxml}{n}[1]{\def\GTS@@drawMetaNot{1}}
481 \define@key{GTSxml}{and}[1]{\def\GTS@@drawMetaAnd{1}}
482 \define@key{GTSxml}{a}[1]{\def\GTS@@drawMetaAnd{1}}
483 \define@key{GTSxml}{or}[1]{\def\GTS@@drawMetaOr{1}}
484 \define@key{GTSxml}{o}[1]{\def\GTS@@drawMetaOr{1}}
485 \define@key{GTSxml}{relation}[1]{\def\GTS@@relation{1}}
486 \define@key{GTSxml}{rel}[1]{\def\GTS@@relation{1}}
487 \define@key{GTSxml}{r}[1]{\def\GTS@@relation{1}}
488
489 \newcommand{GTSxml}[1][]{%
490 \def\GTS@@drawMetaNot{}%
491 \def\GTS@@drawMetaAnd{}%
492 \def\GTS@@drawMetaOr{}%
493 \def\GTS@@relation{}%
494 \setkeys{GTSxml}{#1}%
495 \if\GTS@@drawMetaNot1%
496 \GTSmN/%
497 \else\if\GTS@@drawMetaAnd1%
498 \if\GTS@@relation{1}%
499 \GTSmAR/%
500 \else%
501 \GTSmA/%
502 \fi%
503 \else\if\GTS@@drawMetaOr1%
504 \if\GTS@@relation{1}%
505 \GTSmOR/%
```

```

506 \else%
507 \GTSm0/%
508 \fi%
509 \else%
510 Wrong call to \textbackslash{}GTSxml%
511 \fi%
512 \fi%
513 \fi%
514 }
515
516 \newcommand{\GTSxmetallogic}[1] [] {\GTSxml[#1]}

```

\GTSx One macro to rule them all. Versatile macro for all symbols that can be given options, and a symbol type mandatory switch.

Argument:

[*\optionsList*] Sets the following options if not empty:

- If $\{\langle symbolType \rangle\}$ below is **a**, see \GTSxa:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
up or u,
down or d,
relation or rel or r.
- If $\{\langle symbolType \rangle\}$ below is **i**, see \GTSxi:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
in or i,
out or o,
relation or rel or r.
- If $\{\langle symbolType \rangle\}$ below is **m**, see \GTSxml:
not or n,
and or a,
or or o,
relation or rel or r.

$\{\langle symbolType \rangle\}$ a mandatory switch; it can take value **a** for adjacency, **i** for incidence, **m** for meta-logic.

```

517 \newcommand{\GTSx}[2] [] {%
518 \if#2a%
519 \GTSxa[#1]%
520 \else\if#2i%
521 \GTSxi[#1]%
522 \else\if#2m%
523 \GTSxml[#1]%
524 \else%
525 Wrong call to \textbackslash{}GTSx%
526 \fi%
527 \fi%
528 \fi%
529 }

530 \endinput
531 \</package>

```

4 Acknowledgments

Thank You God! Thank You Father! Thank You Jesus! Thank You Holy-Spirit!

5 Support

This package is gratis and free. But if you can star it on GitHub, it is a nice encouragement.

6 Change History

v0.1.0 – 2026-08-01	v1.3.0 – 2026-09-08
General: Initial version 1	General: New macros for monochrome symbols . . . 1
v1.0.0 – 2026-08-26	v1.4.0 – 2026-09-12
General: First public release 1	General: New macros for symbols for logical
v1.1.0 – 2026-08-29	operations on relations 1
General: Added a memo 1	v1.5.0 – 2026-09-12
v1.1.1 – 2026-08-30	General: One macro to rule them all 1
General: Small corrections and improvements 1	v1.5.1 – 2026-09-13
v1.2.0 – 2026-09-06	General: Use noprint option of doc package to
General: CTAN URL, new macros for same set of	avoid too big left margins, and use paragraph
symbols 1	command instead 1
v1.2.1 – 2026-09-08	
General: Smaller tests output 1	

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